

Bike Sales Analysis in India

Project Objective

- The main objective of this project is to analyze bike sales and resale data in India to understand pricing patterns, depreciation, resale value, vehicle age, and brand performance. The project uses Excel, SQL, and Power BI to clean, process, analyze, and visualize the data.
- The analysis aims to identify important trends and insights that can help understand factors affecting bike resale prices and support better business decision-making.

Project Scope

- This project focuses on analyzing bike sales and resale data in India to identify important patterns and trends in the bike market. The scope includes data cleaning, transformation, analysis, and visualization using Excel, SQL, and Power BI.
- The project covers factors such as original price, resale price, depreciation, resale percentage, vehicle age, brand, fuel type, and other relevant bike attributes. The analysis is presented through KPIs, charts, filters, and an interactive Power BI dashboard to make the findings easy to understand.
- The project also aims to provide useful insights into bike pricing, depreciation, resale performance, and factors that influence resale value.

Tools & Technologies

- The project uses the following tools and technologies for data preparation, analysis, and visualization:
- Microsoft Excel – Used for initial data cleaning, formatting, handling missing values, identifying duplicates, and basic analysis.
- SQL – Used to store, retrieve, filter, aggregate, and analyze the bike dataset using SQL queries.
- Power BI – Used to create KPIs, interactive charts, slicers, dashboard pages, and data visualizations.
- DAX (Data Analysis Expressions) – Used in Power BI to create calculated measures and perform analytical calculations such as average resale price, depreciation, and resale percentage.

Data Description

- The dataset used in this project contains information about bikes available in the Indian market, including details related to their original price, resale price, vehicle age, brand, and other relevant attributes.
- The dataset is used to analyze the relationship between the original purchase price and resale price and to understand how factors such as vehicle age and brand affect resale performance.
- Key fields used in the analysis include:
 - **Brand** – Name of the bike manufacturer or brand.
 - **Model** – Bike model name.
 - **Original Price** – Price of the bike when purchased.
 - **Resale Price** – Price at which the bike is resold.

- **Manufacturing Year** – Year in which the bike was manufactured.
- **Vehicle Age** – Age of the bike calculated from its manufacturing year.
- **Fuel Type** – Type of fuel used by the bike.
- **Depreciation** – Difference between the original price and resale price.
- **Resale Percentage** – Percentage of the original price retained at resale.
- The dataset provides the required information for calculating KPIs and creating visualizations in Power BI.

Data Cleaning & Preprocessing

- Before performing the analysis, the dataset was cleaned and prepared to improve data quality and ensure accurate results.
- The following preprocessing steps were performed:
- **Identified Missing Values** – Checked the dataset for blank or missing values and handled them appropriately.
- **Removed Duplicate Records** – Identified and removed duplicate rows to avoid duplicate information affecting the analysis.
- **Corrected Data Types** – Ensured that numerical fields such as Original Price and Resale Price were stored as numbers and that year-related fields were stored correctly.
- **Standardized Text Values** – Corrected inconsistent values in fields such as Brand and Fuel Type.
- **Handled Incorrect Values** – Identified invalid or inconsistent records and corrected or removed them where necessary.

- Checked Outliers – Examined unusually high or low prices and other numerical values to determine whether they were valid.
- Created Calculated Columns – Created fields such as Vehicle Age, Depreciation, and Resale Percentage for further analysis.
- Validated the Cleaned Data – Checked the final dataset to ensure that the data was consistent and suitable for SQL and Power BI analysis.

KPI requirement:

1. **Average Resale Price:** To understand the typical resale value of bike
2. **Average Depreciation:** To understand how much value bikes lose on average.
3. **Average Resale % :** To measure how much of the original value bikes retain.
4. **Highest Resale Value :** To identify the bike with the highest resale value.
5. **Lowest Resale Value:** To identify the bike with the lowest resale value.
6. **Average Original Price:** To understand the average initial bike price.
7. **Average Vehicle Age:** To understand the typical age of bikes in the dataset.

Chart Requirements:

1. **Average Resale Price by Brand: Bar Chart**-To compare the average resale value of different brands.

2. **Resale Price by Vehicle Age: Line Chart**-To understand how resale value changes as the bike gets older. It shows depreciation behaviour.
3. **Original Price vs Resale Price: Scatter Chart**-To identify the relationship between original price and resale price
4. **Resale Price by State: Bar Chart**-To compare bike resale values across different states in India.
5. **Resale Percentage by Fuel Type: Donut Chart**-To compare value retention across different fuel types.
6. **Resale Price by Owner Type: Column Chart**-To understand how ownership history relates to resale value.
7. **Average Resale Price by City Tier: Column Chart**-To compare resale values across different city tiers.
8. **Top 10 Models by Resale Price: Bar Chart**-To identify models with the highest average resale value.

Formulas Used: For New measure:

1. **To Find Average Resale Price:** $\text{Average_resale_Price} = \text{AVERAGE}(\text{bike_sales_india}[\text{Resale Price (INR)}])$
2. **To Find Average Depreciation:** $\text{Average_Depreciation} = \text{AVERAGE}(\text{bike_sales_india}[\text{DEPRECIATION AMOUNT}])$
3. **To Find Average Resale %:** $\text{Average_Sales\%} = \text{AVERAGE}(\text{bike_sales_india}[\text{Resale Percentage}])$
4. **To Find Highest Resale Value :** $\text{Highest Resale Value} = \text{MAX}(\text{bike_sales_india}[\text{Resale Price (INR)}])$

5. **To Find Lowest Resale Value:** Lowest Resale Value =
 $\text{MIN}(\text{bike_sales_india}[\text{Resale Price (INR)}])$
6. **To Find Average Original Price:** Average Original Price =
 $\text{AVERAGE}(\text{bike_sales_india}[\text{Price (INR)}])$
7. **To Find Average Vehicle Age:** Average Vehicle Age =
 $\text{AVERAGE}(\text{bike_sales_india}[\text{Vehicle age}])$

Formulas Used: For New Columns:

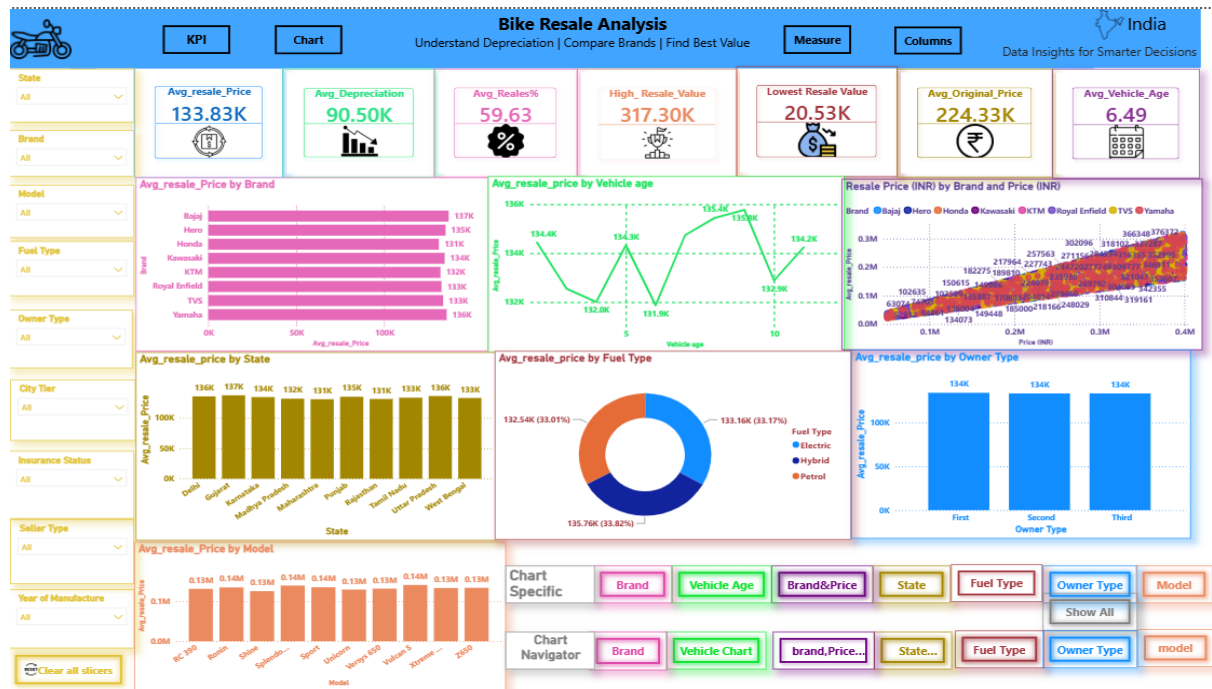
1. **Vehicle Age:** Vehicle age = $\text{YEAR}(\text{TODAY}()) - [\text{Year of Manufacture}]$
2. **DEPRECIATION AMOUNT :** $\text{DEPRECIATION AMOUNT} = [\text{Price (INR)}] - [\text{Resale Price (INR)}]$
3. **depreciation percentage :** $\text{depreciation percentage} = \frac{\text{bike_sales_india}[\text{Price (INR)}] - \text{bike_sales_india}[\text{Resale Price (INR)}]}{\text{bike_sales_india}[\text{Price (INR)}]} * 100$
4. **Resale Percentage :** $\text{Resale Percentage} = \frac{\text{DIVIDE}(\text{bike_sales_india}[\text{Resale Price (INR)}], \text{bike_sales_india}[\text{Price (INR)}], 0)}{\text{bike_sales_india}[\text{Price (INR)}]} * 100$

SQL Analysis

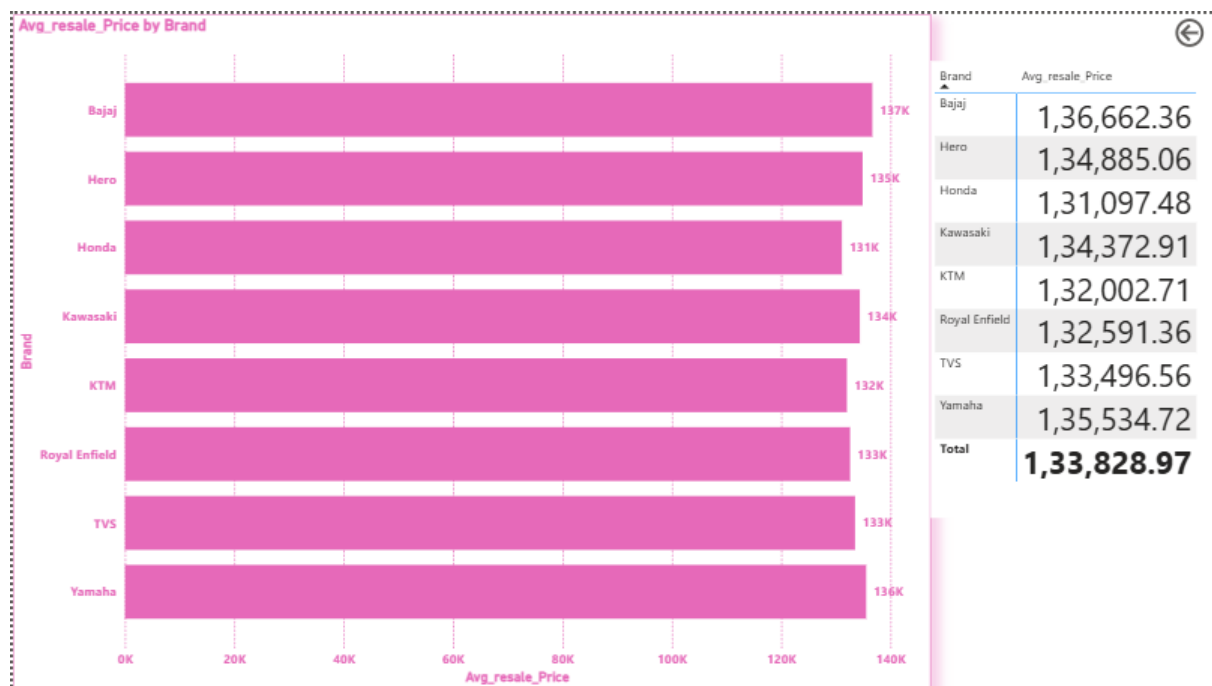
- SQL was used to store, retrieve, clean, and analyze the bike sales dataset. The cleaned dataset was imported into a SQL database, where queries were used to perform calculations, filtering, grouping, and comparisons.
- The following SQL analysis was performed:

- **Data Retrieval:** Retrieved bike records from the database using SELECT queries. Selected specific columns required for analysis.
- **Data Filtering:** Filtered records based on brand, fuel type, vehicle age, price range, and other conditions using WHERE.
- **Aggregation:** Calculated metrics such as average, minimum, maximum, and total values using AVG(), MIN(), MAX(), and SUM().
- **Brand-wise Analysis:** Compared the average original price and resale price of different bike brands using GROUP BY.
- **Fuel Type Analysis:** Analyzed bike count and pricing based on different fuel types.
- **Resale Price Analysis:** Compared original price and resale price to understand the resale performance of bikes.
- **Depreciation Analysis:** Calculated depreciation using the difference between the original price and resale price.
- **Vehicle Age Analysis:** Analyzed how vehicle age is related to resale price and depreciation.
- **Sorting and Ranking:** Used ORDER BY to identify bikes with the highest and lowest resale prices.
- **Data Validation:** Used SQL queries to identify missing, duplicate, or inconsistent records before using the data in Power BI.
- The results obtained from SQL analysis were used as a foundation for creating KPIs, charts, and interactive dashboards in Power BI.

Bike Sales In India Analysis Dashboard

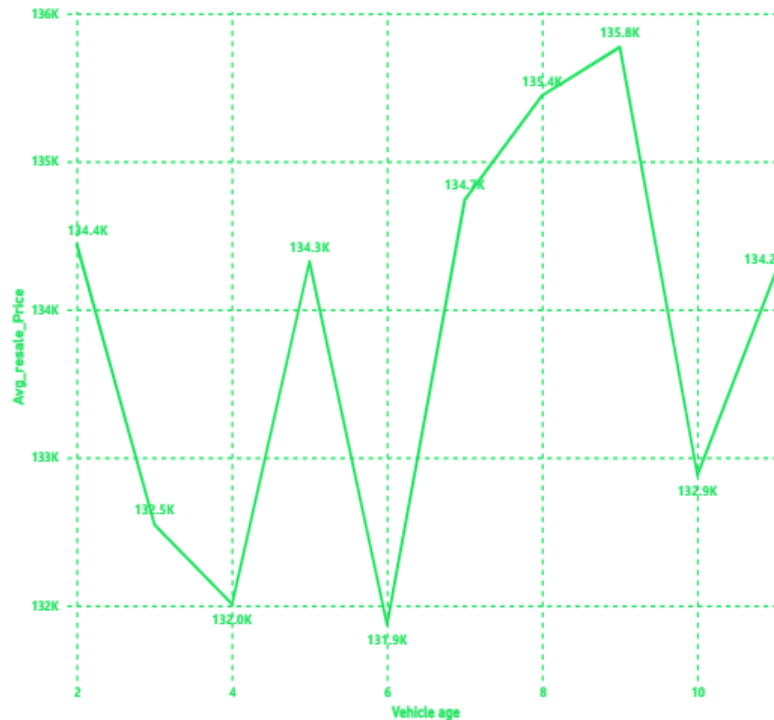


Brand Analysis



Vehicle Age Analysis

Avg_resale_price by Vehicle age

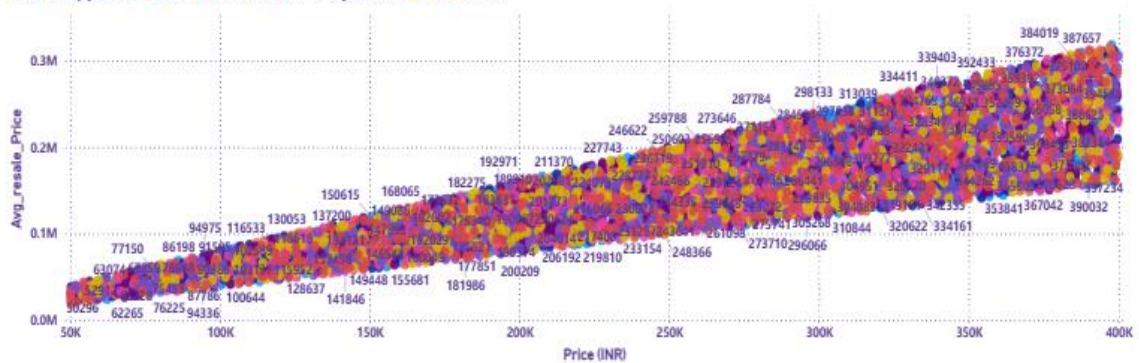


Vehicle age	Avg resale Price
2	1,34,439.73
3	1,32,547.34
4	1,32,011.21
5	1,34,324.95
6	1,31,873.34
7	1,34,743.03
8	1,35,447.46
9	1,35,774.08
10	1,32,882.75
11	1,34,240.93
Total	1,33,828.97

Brand And price Analysis

Resale Price (INR) by Brand and Price (INR)

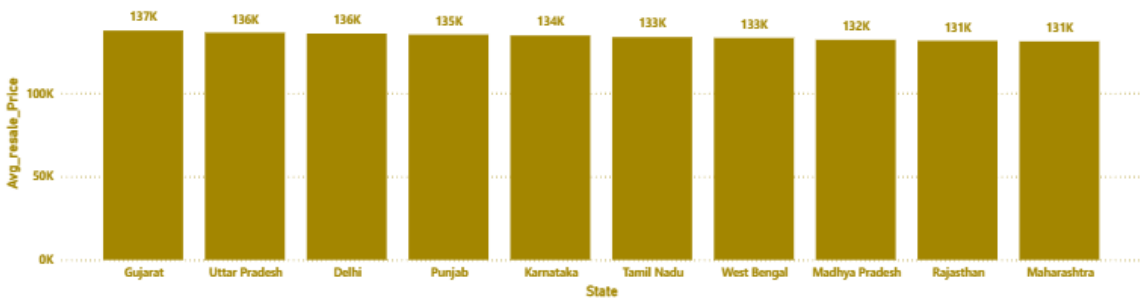
Brand Bajaj Hero Honda Kawasaki KTM Royal Enfield TVS Yamaha



Price (INR)	Bajaj	Hero	Honda	Kawasaki	KTM	Royal Enfield	TVS	Yamaha	Total
50031		34,348.76							34,348.76
50074						21,360.26			21,360.26
50112			39,410.34						39,410.34
50141	36,559.21							30,222.17	33,390.69
50158								26,153.68	26,153.68
50169			26,040.86						26,040.86
50187						30,535.86			30,535.86
50255				35,798.05					35,798.05
50263								20,532.07	20,532.07
50277			33,983.42						33,983.42
50296		32,095.15							32,095.15
50299						35,572.68			35,572.68
50303								31,587.02	31,587.02
Total	1,36,662.36	1,34,885.06	1,31,097.48	1,34,372.91	1,32,002.71	1,32,591.36	1,33,496.56	1,35,534.72	1,33,828.97

State Analysis

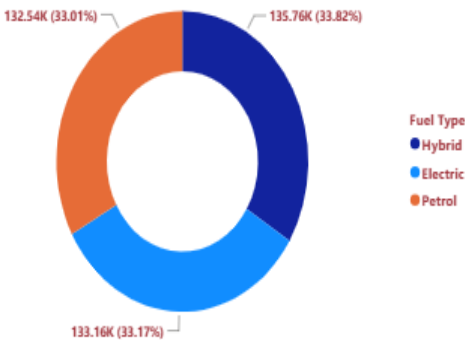
Avg_resale_price1 by State



Brand	Delhi	Gujarat	Karnataka	Madhya Pradesh	Maharashtra	Punjab	Rajasthan	Tamil Nadu	Uttar Pradesh	West B
Bajaj	1,47,209.18	1,49,395.33	1,36,337.53	1,36,942.91	1,26,569.14	1,39,064.92	1,34,880.59	1,24,631.57	1,35,934.61	1,35,934.61
Hero	1,34,813.44	1,29,952.55	1,42,818.94	1,36,639.59	1,45,159.22	1,39,783.08	1,32,977.69	1,31,965.49	1,29,893.50	1,29,893.50
Honda	1,29,928.95	1,32,076.84	1,30,900.54	1,32,021.04	1,29,909.57	1,24,405.38	1,24,321.22	1,39,303.58	1,35,914.39	1,35,914.39
Kawasaki	1,44,840.69	1,31,785.49	1,35,841.56	1,30,660.53	1,28,467.34	1,39,630.52	1,28,212.86	1,32,973.32	1,41,385.96	1,41,385.96
KTM	1,25,974.55	1,32,903.35	1,31,046.81	1,36,410.48	1,28,390.37	1,35,688.22	1,35,980.06	1,29,787.01	1,24,321.99	1,24,321.99
Royal Enfield	1,34,891.34	1,46,773.26	1,29,173.50	1,27,978.42	1,20,836.26	1,30,923.44	1,33,146.41	1,30,141.41	1,33,384.94	1,33,384.94
TVS	1,30,544.93	1,46,475.05	1,37,908.69	1,26,189.95	1,34,082.74	1,29,186.59	1,37,087.28	1,34,757.65	1,32,103.20	1,32,103.20
Yamaha	1,38,773.13	1,31,140.31	1,31,092.44	1,29,509.37	1,34,853.13	1,41,261.54	1,21,120.87	1,41,460.34	1,56,251.58	1,56,251.58
Total	1,35,509.97	1,37,316.16	1,34,381.21	1,31,771.14	1,30,839.17	1,34,903.60	1,31,219.33	1,33,388.84	1,36,084.87	1,36,084.87

Fuel Type Analysis

Avg_resale_price by Fuel Type



Brand	Electric	Hybrid	Petrol	Total
Bajaj	1,35,555.17	1,39,020.70	1,35,423.34	1,36,662.36
Hero	1,36,071.29	1,39,589.43	1,28,840.20	1,34,885.06
Honda	1,33,326.84	1,29,865.69	1,30,191.56	1,31,097.48
Kawasaki	1,34,365.94	1,33,734.65	1,34,968.00	1,34,372.91
KTM	1,29,906.96	1,37,115.15	1,28,881.60	1,32,002.71
Royal Enfield	1,28,859.23	1,34,092.68	1,34,420.40	1,32,591.36
TVS	1,33,274.19	1,35,749.61	1,31,142.15	1,33,496.56
Yamaha	1,33,669.05	1,36,806.70	1,36,035.24	1,35,534.72
Total	1,33,162.12	1,35,764.86	1,32,543.52	1,33,828.97

Owner Type Analysis



Model Analysis



Key Insights

- The analysis provides insights into bike pricing and resale performance in the Indian market.
- The major areas identified through the analysis include:
- **Resale Price Analysis** – Comparison of original prices and resale prices helps identify how much value bikes retain over time.
- **Depreciation Analysis** – The difference between original price and resale price helps understand the depreciation of bikes.
- **Vehicle Age Impact** – Analysis of vehicle age helps identify the relationship between the age of a bike and its resale value.
- **Brand Performance** – Comparing different brands helps identify variations in average original price, resale price, and resale percentage.
- **Fuel Type Analysis** – Comparing fuel types provides an understanding of differences in pricing and resale performance.
- **Resale Percentage** – Resale percentage shows how much of the original bike value is retained at the time of resale.
- **Price Comparison** – Comparing original and resale prices provides a clear view of the overall pricing pattern in the dataset.
- These insights are presented through KPIs, charts, slicers, and interactive Power BI dashboard pages.

Conclusion

- The Bike Sales Analysis project demonstrates how data analytics tools can be used to understand bike pricing and resale performance in India. The dataset was cleaned and processed

using Excel and SQL, and the results were analyzed and visualized using Power BI.

- The project provides a clear view of important metrics such as average resale price, depreciation, resale percentage, original price, and vehicle age. Interactive dashboards and visualizations make it easier to identify patterns and compare bike resale performance across different categories.
- Overall, the project demonstrates the complete data analytics workflow, from data cleaning and SQL analysis to KPI creation, visualization, and business insights using Excel, SQL, and Power BI.

